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 Studierichting: Informatica
 Oefeningenreeks 1D nr. 20.1

$$20.1) 4z^4 - 16z^2 - 64 = 0$$

$$\begin{array}{c|ccccc} 2 & 4 & 0 & -15 & 0 & -4 \\ & & 8 & 16 & 2 & 4 \\ \hline & 4 & 8 & 1 & 2 & 0 \end{array}$$

$$(z - 2)(4z^3 + 8z^2 + z + 2)$$

$$\begin{array}{c|cccc} -2 & 4 & 8 & 1 & 2 \\ & & -8 & 0 & -2 \\ \hline & 4 & 0 & 1 & 0 \end{array}$$

$$(z - 2)(z + 2)(4z^2 + 1) = 0$$

$$\Rightarrow 4z^2 + 1 = 0$$

$$z^2 = -\frac{1}{4}$$

$$z = \pm \frac{1}{2}i$$

Nulpunten:

- $z_1 = 2$
- $z_2 = -2$
- $z_3 = \frac{1}{2}i$
- $z_4 = -\frac{1}{2}i$

$$(z - 2)(z + 2)(4z^2 + 1) = 0$$

References